

Mark Christian Messner

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Education

2011-2014	Doctor of Philosophy , <i>University of Illinois at Urbana-Champaign</i> GPA 3.96/4.00 Major: Civil and Environmental Engineering Advisor: Robert Dodds, Jr. Dissertation: <i>Micromechanical models of delamination in Al-Li alloys</i> Computational Science and Engineering Certificate
2010-2011	Master of Science , <i>University of Illinois at Urbana-Champaign</i> GPA 3.96/4.00 Major: Civil and Environmental Engineering Advisor: Robert Dodds, Jr. Computational Science and Engineering Certificate
2006-2010	Bachelor of Science , <i>University of Illinois at Urbana-Champaign</i> GPA 3.97/4.00 Major: Civil and Environmental Engineering, Minor: German Degree awarded with Highest Honors and University Honors

Appointments

2023-	Group Leader , <i>Argonne National Laboratory</i> Established the Thermal and Structural Materials Modeling and Simulations Group. Led a team of 6 staff and 4-6 postdocs on projects funded by the U.S. Department of Energy, Office of Nuclear Energy, the Office of Critical Minerals and Energy Innovation, the Advanced Materials and Manufacturing Technologies Office, the National Nuclear Security Administration, the U.S. Nuclear Regulatory Agency, and other sponsors. Managed a research portfolio with greater than \$3 million per year in total funding. Established new research portfolios in modeling and simulation for advanced manufacturing processes and the reliability of monolithic ceramics and ceramic matrix composites. Recognized as a national expert in high temperature design.
2016-	Principal Mechanical Engineer , <i>Argonne National Laboratory</i> Research topics: High temperature structural materials, design of high temperature nuclear reactors and concentrating solar power systems, crystal plasticity, machine learning methods for materials and material constitutive modeling, qualification of AM nuclear components Led work on the revision and improvement of several parts of the American Society of Mechanical Engineers Boiler and Pressure Vessel Code, Section III, Division 5 covering the design and construction of high temperature nuclear reactors
2014-2016	Postdoctoral Researcher , <i>Lawrence Livermore National Laboratory</i>

Supervisor: Nathan Barton
 Research topics: Multiscale material modeling of additively manufactured structured materials, modeling and optimization of lattice-structured meta-materials, multiscale modeling of HCP metals

2010-2014 **Research Assistant**, *University of Illinois at Urbana-Champaign*
 Supervisor: Robert Dodds, Jr.
 Research topics: Parallel performance of WARP3D, crystal plasticity, mesoscale modeling of fatigue/fracture processes, homogenization and multiscale damage calculations

Honors/Awards

2025	ASME G.E.O. Widera Literature Award for Authoring the Outstanding Technical Paper in the Journal of Pressure Vessel Technology
2023	Doug Scarth Early Career Leadership Award for Outstanding Service to the Pressure Vessels & Piping (PVP) Division as an Early Career Engineer
2022	Secretary of Energy Achievement Award
2022, 2023, 2025	ASME Boiler & Pressure Vessel Code Certificate of Acclamation
2020	Impact Argonne Award for Innovation
2012-14	National Defense Science & Engineering Graduate Fellowship
2010-11	University Fellowship
2008-	Tau Beta Pi

Professional Affiliations

2016-	ASME
2016-	ANS
2014-2016	APS
2006-2014	ASCE, EMI

Professional Service

2018-	Generation IV Forum: Task Group/Working Group on Advanced Manufacturing and Materials Engineering <i>co-chair</i>
2020-	ASME Boiler & Pressure Vessel Code <i>committee chair</i> : BPV III WG on Analysis Methods
2018-2020	ASME Boiler & Pressure Vessel Code <i>committee chair</i> : BPV III SWG on Inelastic Analysis Methods
2017-	ASME Boiler & Pressure Vessel Code <i>committee member</i> : BPV III SG High Temperature Reactors, WG on Analysis Methods, WG High Temperature Flaw Evaluation, WG Creep-Fatigue and Negligible Creep; BPTCS/BNCS Special Committee on Use of Additive Manufacturing for Pressure Retaining Equipment, BVP I/VIII WG on Elevated Temperature Design, and several other committees.
2017-	PVP Conference: Technical Program Representative, co-Technical Program Representative, Honors and Awards Chair, Materials and Fabrication Secretary, Vice Chair, and Chair, track co-chair
2018-2019	WCCM/USNCCM: <i>track organizer</i>
2026	<i>Reviewer for (past year)</i> : Fatigue and Fracture of Engineering Materials and Structures, Nuclear Science and Engineering, Meccanica, Integrating Materials and Manufacturing Innovation, International Journal for Numerical Methods in Engineering, International Journal of Fatigue, Mechanics of Materials, Journal of Alloys and Compounds, International Journal of Pressure Vessels and Piping

	ASME Journal of Pressure Vessel Technology, and ASME PVP Conference Proceedings
2018-	External proposal reviewer for DOE:NE, DOE:EERE, DOE:FES, the NSF, the German Research Foundation, and the Dutch Research Council Domain Applied and Engineering Sciences

Institutional and Community Service

2026-	Argonne AI Governance Council WG - AI Agent Lifecycle Management
2019-	Library User Committee member
2015-	Volunteer at middle school/high school DOE Science Bowl
2020-	STEM chat volunteer for local elementary and high schools

PhD Committee Service

2018-2021	Alon Katz (Georgia Institute of Technology)
2022-2025	Janzen Choi (University of New South Wales)
2024-	Jessie Weng May Lum (University of New South Wales)
2025-	Milly Yi (Carnegie Mellon University)
2025-	Colin Gerlach (University of Miami)

DOE:NE Work Packages Managed

2022-	AMMT: Several work packages – \$700k/year
2021-	ART: Several work packages – \$700k/year
2019-	NEAMS: Structural Materials – \$500k/year
2020-2021	NDMQi: High Temperature Qualification – \$200k/year

Funding Awards as PI

2026-2027	NNSA AI4NS: Foundation models for structured composite material discovery – \$8M (\$4M ANL and \$4M collaborator funding)
2024-2027	DOE:EERE: Validated time-dependent reliability modeling for high temperature CSP systems – \$2.5M
2024-2025	DOE:AMMTO: PUMA (Powder Utilization Modeling Application): a MOOSE-based application for modeling powder post-processing for advanced manufacturing – \$500k
2024-2025	US NRC: US NRC: Risk Informed Comparison of ASME Section III, Division 5 and ASME Section VIII, Divisions 1 and 2 – \$100k
2025-2026	EPRI: Accelerated Qualification of 316L for ASME Section III, Division 5, Class A Construction – \$80k
2024-2024	EPRI: Accelerating qualification of structural materials for advanced reactors – \$80k
2023-2024	EPRI: Accelerating qualification of structural materials for advanced reactors – \$80k
2023-2024	US NRC: Technical Assessment of ASME BPVC Section III, Div. 5 Composites Rules – \$360k
2022-2024	Argonne Laboratory Directed Research and Development (LDRD): Microarchitected Composites – \$300k
2022-2024	US NRC: Technical Assistance Pertaining to Advanced Reactors - Assessment of Salt Properties, Stress Relaxation Cracking and Materials/Component Integrity – \$150k
2021-2022	DOE:HPC4Energy : An ICME Modeling Framework for Metal Matrix Composites

	Focusing on Ultrahigh Temperature Matrix Material with Tungsten Carbide Reinforcement – \$300k
2021-2023	DOE:EERE: Design Methods, Tools, and Data for Ceramic Solar Receivers – \$955k
2020-2021	DOE:EERE : High Temperature Receiver Design Package – \$517k
2019-2020	US NRC : Assess State of Knowledge of Modeling and Simulation and Microstructural Analysis for Advanced Manufacturing Technologies (AMTs) – \$200k
2019-2021	DOE:NE FOA: Modeling and Simulation Development Pathways to Accelerate KP-FHR Licensing (topic PI) – \$500k
2018-2020	DOE:EERE: Creep-fatigue design for CSP receivers (topic PI) – \$375k
2016	LLNL TechBase: Adaptive smart materials – \$65k
2015	LLNL TechBase: Material model library for lattice structured meta-materials – \$50k

Other Skills and Qualifications

Languages: German (Proficient)

Publications/Presentations

Refereed journal publications

- [1] Bipul Barua et al. “Design of a SiC-Si moving packed-bed particle-to-sCO₂ heat exchanger for high temperature concentrating solar power applications”. In: *Solar Energy* 303 (2026), p. 114114.
- [2] Shahmeer Baweja and Mark C Messner. “Development of predictive model for accurate rupture time from multi-axial creep in alloy 709 with physics-based simulations”. In: *Computational Materials Science* 263 (2026), p. 114427.
- [3] Sagar Bhatt and Mark C. Messner. “Modeling the Impact of Grain Morphology and Porosity on Creep Anisotropy in Laser Powder Bed Fusion Materials”. In: *Integrating Materials and Manufacturing Innovation* 15 (2026), pp. 649–662.
- [4] Abhishek S Bhesania et al. “The role of moisture in MgCl₂ salt: A multiscale approach to TES performance”. In: *International Journal of Heat and Mass Transfer* 256 (2026), p. 128015.
- [5] Tianju Chen, Mark Christian Messner, and Huadon Fu. “Hierarchical structured Stochastic Variational Inference for uncertainty quantification of viscoplastic constitutive models”. In: *European Journal of Mechanics: A / Solids* (2026).
- [6] Hao Deng and Mark C. Messner. “Data-Driven Method for Modeling Creep-Fatigue Stress-Strain Behavior Using Neural ODEs”. In: *Optimization and Engineering* 27 (2026), pp. 617–650.
- [7] Hao Deng and Mark C. Messner. “Metamaterial Design based on Autoencoder Representation using an Active Learning Method”. In: *Structural and Multidisciplinary Optimization* 69.1 (2026), p. 21.
- [8] Wenchao Du et al. “Micro-architected Metallic Metamaterial Composites by Data-driven Generative Design and Binder Jetting Additive Manufacturing”. In: *Additive Manufacturing* (2026). in press.
- [9] Tianchen Hu et al. “Version 3—NEML2: An efficient and modular multiphysics constitutive modeling library for hybrid computing environments”. In: *SoftwareX* 35 (2026).
- [10] J. Lum et al. “Comparative Creep–Fatigue Assessment of a Representative Intermediate Heat Exchanger in a High-Temperature Gas-Cooled Reactor System”. In: *Nuclear Engineering and Technology* (2026).
- [11] M. Petkov, M. Messner, and M. Spindler. “Assessing the Accuracy of Time-Fraction and Ductility Exhaustion Approaches for Creep-Fatigue Damage Prediction in Alloy 617 Through Feature Test Validation”. In: *Fatigue & Fracture of Engineering Materials & Structures* (2026).

- [12] Huy Tran et al. “PUMA: A scalable framework for simulating powder post-processing in advanced manufacturing”. In: *Advances in Engineering Software* 219 (2026).
- [13] Cheng-Hau Yang, Mark C. Messner, and Tianchen Hu. “A Shifted Cohesive-Zone Method for Non-Interface-Fitted Meshes with Applications to Crystal Plasticity”. In: *Computer Methods in Applied Mechanics and Engineering* 461 (2026), p. 119259.
- [14] Bipul Barua, Cody Hale, and Mark C Messner. “Structural Design and Modeling of MARVEL Primary Coolant System Using the ASME Section III, Division 5, Code”. In: *Nuclear Technology* (2025), pp. 1–18.
- [15] Pawan Chaugule et al. “srlife: A software tool for estimating the life of high temperature concentrating solar receivers. Part II–Ceramic receivers”. In: *Solar Energy* 301 (2025), p. 113899.
- [16] J Choi et al. “Multi-objective surrogate-assisted calibration of CPFEM models using macroscopic response and in situ EBSD measurements of grain reorientation trajectories”. In: *Acta Materialia* (2025), p. 121809.
- [17] Tianchen Hu and Mark C Messner. “A Simple, Scalable Large Deformation Solid Mechanics Implementation in the MOOSE Framework”. In: *ACM Transactions on Mathematical Software* 51.1 (2025), pp. 1–22.
- [18] Tianchen Hu and Mark C Messner. “NEML2: An efficient and modular multiphysics constitutive modeling library for hybrid computing environments”. In: *SoftwareX* 31 (2025), p. 102302.
- [19] Chaofan Huang et al. “Calibration of RAFM Micromechanical Model for Creep Using Bayesian Optimization for Functional Output”. In: *Journal of Computing and Information Science in Engineering* 25.3 (2025), p. 031007.
- [20] W Ji et al. “Development, Validation, and Verification of Multi-Pass Thermo-Mechanical Welding Simulations Using the Open-Source MOOSE Framework: NeT TG4 Benchmark Weldment”. In: *International Journal of Pressure Vessels and Piping* (2025), p. 105560.
- [21] Mark C Messner and Bipul Barua. “srlife: a software tool for estimating the life of high temperature concentrating solar receivers. Part I–metallic receivers”. In: *Solar Energy* 295 (2025), p. 113518.
- [22] Mark C Messner and Tianchen Hu. “Fully implicit crystal plasticity models representing orientations with modified Rodrigues parameters”. In: *Mechanics of Materials* (2025), p. 105388.
- [23] Sashank Shivakumar et al. “Current Activated Reactive Ultrafast Joining (CARUJ) of Silicon Carbide”. In: *Ceramics International* (2025).
- [24] Rui Wang et al. “Phase-Field Modeling of Diffusion Bonding in 316H Stainless Steel: Impact of Processing Conditions on Grain Morphology and Bonding Quality”. In: *Materials Science and Engineering: A* (2025), p. 149051.
- [25] Jacob Wenner, Mark C Messner, and Michael J Wagner. “Damage modeling of power tower receiver tubes using the SRLIFE tool”. In: *Solar Energy* 299 (2025), p. 113627.
- [26] Xuan Zhang et al. “Long-term Thermal Aging Behavior and Strength Reduction in a Laser Powder Bed Fusion 316H Stainless Steel”. In: *Acta Materialia* (2025), p. 121491.
- [27] Pawan Chaugule et al. “Reliability comparisons between additively manufactured and conventional SiC-Si ceramic composites”. In: *Journal of the American Ceramic Society* 107.5 (2024), pp. 3117–3133.
- [28] Ezra Mengiste et al. “Effect of irradiation-induced strength anisotropy on the reorientation trajectories and fragmentation behavior of grains in BCC polycrystals under tensile loading”. In: *Acta Materialia* 263 (2024), p. 119503.
- [29] Markian Petkov and Mark Messner. “Application of C (t)-Integral Solutions in Extending ASME BPVC Section XI Division 2 Code Case N-934 for Transient Creep Crack Growth”. In: *Journal of Pressure Vessel Technology* 146.5 (2024).
- [30] B. Barua et al. “Designing Cladded Components for High Temperature Nuclear Service: Part II – Design Rules”. In: *Journal of Pressure Vessel Technology* 145.2 (2023), p. 021302.

- [31] Bipul Barua and Mark C. Messner. “Structural Design Challenges and Implications for High Temperature Concentrating Solar Power Receivers”. In: *Solar Energy* 251 (2023), pp. 119–133.
- [32] Amirfarzad Behnam et al. “Uncertainty Quantification Framework for Predicting Material Response with Large Number of Parameters: Application to Creep Prediction in Ferritic-Martensitic Steels using Combined Crystal Plasticity and Grain Boundary Models”. In: *Integrating Materials and Manufacturing Innovation* 11 (2023), pp. 516–531.
- [33] Tianju Chen and Mark C. Messner. “Training material models using gradient descent algorithms”. In: *International Journal of Plasticity* 165 (2023), p. 103605.
- [34] Tianchen Hu et al. “A Three-Dimensional, Thermodynamically and Variationally Consistent, Fully Coupled, Electro-Chemo-Thermo-Mechanical Model of Solid-State Batteries”. In: *Journal of The Electrochemical Society* 170.12 (2023), p. 123501.
- [35] M. C. Messner et al. “Designing Cladded Components for High Temperature Nuclear Service: Part I – Analysis Methods”. In: *Journal of Pressure Vessel Technology* 145.2 (2023), p. 021301.
- [36] Mark Messner, Guosheng Ye, and T.-L. Sham. “A Structural Design Approach Tailored for the Rapid, Preliminary Design of Microreactor Components”. In: *Nuclear Technology* 209.sup1 (2023), S60–S72.
- [37] Mark C Messner, Tianchen Hu, and Tianju Chen. “Time-vectorized numerical integration for systems of ODEs”. In: *arXiv preprint arXiv:2310.08649* (2023).
- [38] Holly D. Carlton et al. “Incorporating defects into model predictions of metal lattice-structured materials”. In: *Materials Science and Engineering: A* 832 (2022), p. 142427.
- [39] A. M. Katz, Davesh Ranjan, and M. C. Messner. “A Novel Approach for Bounding the Stress Experienced by the Core of Utility-Scale Printed Circuit Heat Exchangers Under Thermohydraulic Loads”. In: *Journal of Pressure Vessel Technology* 144.4 (2022).
- [40] Aritra Chakraborty and Mark C. Messner. “Bayesian analysis for estimating statistical parameter distributions of elasto-viscoplastic material models”. In: *Probabilistic Engineering Mechanics* 66 (2021), p. 103153.
- [41] Aritra Chakraborty, Mark C. Messner, and T.-L. Sham. “A minimum creep rate for 2-1/4Cr-1Mo steel consistent with the ASME Section III, Division 5 rules”. In: *Journal of Pressure Vessel Technology* 134.4 (2021), p. 044502.
- [42] A. M. Katz, M. Messner, and Davesh Ranjan. “A novel approach for bounding the stress experienced by the core of utility-scale printed circuit heat exchangers under thermohydraulic loads”. In: *Journal of Pressure Vessel Technology* (2021).
- [43] Andrea Nicolas, Mark Messner, and T.-L. Sham. “A method for predicting failure statistics for steady state elevated temperature components”. In: *International Journal of Pressure Vessels and Piping* 192 (2021), p. 104363.
- [44] Andrea Rovinelli et al. “Accurate effective stress measures: Predicting creep life for 3D stresses using 2D and 1D creep rupture simulations and data”. In: *Integrating Materials and Manufacturing Innovation* (2021).
- [45] M. C. Messner. “Convolutional neural network surrogate models for the mechanical properties of periodic structures”. In: *Journal of Mechanical Design* 142.2 (2020).
- [46] Dileep Singh et al. “One piece ceramic heat exchanger for concentrating solar power electric plants”. In: *Renewable Energy* 160 (2020), pp. 1308–1315.
- [47] M. C. Messner, V.-T. Phan, and T.-L. Sham. “Evaluating and modeling rate sensitivity in advanced reactor structural materials: 316H, Gr. 91, and A617”. In: *International Journal of Pressure Vessels and Piping* 178 (2019), p. 103997.
- [48] M. C. Messner et al. “A Method for Including Diffusive Effects in Texture Evolution”. In: *Journal of the Mechanics and Physics of Solids* 125 (2019), pp. 785–804.
- [49] M. C. Messner et al. “Combined Crystal Plasticity and Grain Boundary Modeling of Creep in Ferritic-Martensitic Steels, Part 2: The Effect of Stress and Temperature on Engineering and Microstructural Properties”. In: *Modelling and Simulation in Materials Science and Engineering* 27.7 (2019), p. 075010.

- [50] Omar Nassif et al. “Combined Crystal Plasticity and Grain Boundary Modeling of Creep in Ferritic-Martensitic Steels, Part 1: Theory and Implementation”. In: *Modelling and Simulation in Materials Science and Engineering* 27.7 (2019), p. 075009.
- [51] Julie A Jackson et al. “Field responsive mechanical metamaterials”. In: *Science Advances* 4.12 (2018), eaau6419.
- [52] H. D. Carlton et al. “Mapping local deformation behavior in single cell metal lattice structures”. In: *Acta Materialia* 129 (2017), pp. 239–250.
- [53] M. C. Messner et al. “A crystal plasticity model for slip resistance and junction formation in HCP metals”. In: *Modelling and Simulation in Materials Science and Engineering* 25.4 (2017), p. 044001.
- [54] Mark C Messner. “A fast, efficient direct slicing method for slender member structures”. In: *Additive Manufacturing* 18 (2017), pp. 213–220.
- [55] J. A. Hawreliak et al. “Dynamic Behavior of Engineered Lattice Materials”. In: *Scientific Reports* 6 (2016).
- [56] M. C. Messner. “Optimal lattice-structured materials”. In: *Journal of the Mechanics and Physics of Solids* 96 (2016), pp. 162–183.
- [57] M. C. Messner, A. J. Beaudoin, and R. H. Dodds, Jr. “A grain boundary damage model for delamination”. In: *Computational Mechanics* 56 (2015), pp. 1–20.
- [58] M. C. Messner, R. H. Dodds, Jr., and A. J. Beaudoin. “Consistent crystal plasticity kinematics and linearization for the implicit finite element method”. In: *Engineering Computations* 32.6 (2015), pp. 1526–1548.
- [59] M. C. Messner et al. “Wave propagation in equivalent continua representing truss lattice materials”. In: *International Journal of Solids and Structures* 73-74 (2015), pp. 55–66.
- [60] M.C. Messner, A. J. Beaudoin, and R. H. Dodds, Jr. “An interface compatibility/equilibrium mechanism for delamination fracture in aluminum-lithium alloys”. In: *Engineering Fracture Mechanics* 133 (2015), pp. 70–84.
- [61] M.C. Messner, A. J. Beaudoin, and R. H. Dodds, Jr. “Mesoscopic modeling of crack arrestor delamination in Al-Li: Primary crack shielding and T-stress effect”. In: *International Journal of Fracture* 188.2 (2014), pp. 229–249.

Pending refereed journal publications

Refereed conference publications

- [62] Bipul Barua and Mark Messner. “Assessment of ASME Section III, Division 5 Rules for Ceramic Matrix Composites in High-Temperature Nuclear Reactor Applications”. In: *Proceedings of the ASME 2026 Pressure Vessels & Piping Conference*. 2026.
- [63] Bipul Barua et al. “Application of Simplified Methods to Size a Surveillance Article and Estimate Remaining Life: A Sample Problem”. In: *Proceedings of the ASME 2026 Pressure Vessels & Piping Conference*. 2026.
- [64] Heramb Mahajan et al. “Alloy 625 Qualification for Elevated Temperature Nuclear Components”. In: *Proceedings of the ASME 2026 Pressure Vessels & Piping Conference*. 2026.
- [65] Caleb Massey et al. “Qualifying Nuclear Structural Materials with Sparse Data: A Gap Analysis for Irradiation Effects Experiments and Facilities”. In: *Proceedings of the ASME 2026 Pressure Vessels & Piping Conference*. 2026.
- [66] Mark Messner et al. “An outline for qualifying laser-powder bed fusion 316 for Section III, Division 5, Class A construction”. In: *Proceedings of the ASME 2026 Pressure Vessels & Piping Conference*. 2026.
- [67] Markian Petkov, Mark Messner, and M. Spindler. “Development and Validation of a Ductility-Exhaustion-Based Creep-Fatigue Evaluation Route for ASME BPVC Section III Division 5”. In: *Proceedings of the ASME 2026 Pressure Vessels & Piping Conference*. 2026.

- [68] Bipul Barua, Tinchun Hu, and Mark C. Messner. “Guidance on Treatment of Welding Residual Stresses in Design Evaluations Using Section III, Division 5 Rules”. In: *ASME 2025 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2025.
- [69] B. W. Brust et al. “High Temperature Flaw Evaluation Code Case N-934: Effect of Transients”. In: *ASME 2025 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2025.
- [70] L. Ivan et al. “A Case Study Assessment of Qualification Acceleration Informed by Physics-Based Modelling”. In: *2025 Regulatory Frameworks and Technical Approaches for Qualification and Through-Life Performance of Materials in Advanced Reactors Workshop*. Nuclear Energy Agency. 2025.
- [71] Michael McMurtrey et al. “Approach for Qualifying Laser Powder Bed Fusion 316H in Section III, Division 5”. In: *ASME 2025 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2025.
- [72] Mark C. Messner and Bipul Barua. “Class A Design Data for Alloy 709 for Use With the ASME Boiler and Pressure Vessel Code Section III, Division 5 Rules”. In: *ASME 2025 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2025.
- [73] Mark C. Messner, Guosheng Ye, and Bipul Barua. “Material Surveillance Concepts and Acceptance Criteria for Future High Temperature Reactors”. In: *ASME 2025 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2025.
- [74] Mark C. Messner et al. “Qualification of Laser Powder Bed Fusion 316H Stainless Steel for High Temperature Reactor Components”. In: *2025 Regulatory Frameworks and Technical Approaches for Qualification and Through-Life Performance of Materials in Advanced Reactors Workshop*. Nuclear Energy Agency. 2025.
- [75] Guosheng Ye, Mark C. Messner, and Bipul Barua. “Simplified Methods for Damage Inference Based on Out-Reactor Test Results”. In: *ASME 2025 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2025.
- [76] Sagar Bhatt et al. “Microstructural Models for the Creep Strength and Ductility of Diffusion-Bonded 316H Steel”. In: *10th International Conference on Advances in Materials, Manufacturing and Repair for Power Plants*. 2024.
- [77] M. C. Messner. “A Standard Form for Cyclic Plasticity Models Used with the ASME Section III, Division 5 Rules”. In: *ASME 2024 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2024.
- [78] Bipul Barua and Mark C Messner. “A Comprehensive Sample Problem for Section III Division 5 Design by Elastic Analysis”. In: *ASME 2023 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2023.
- [79] Bipul Barua et al. “Design Data for Alloy 740H High Temperature Concentrating Solar Power Components”. In: *ASME 2023 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2023.
- [80] Bipul Barua et al. “Time-dependent Failure Assessment of Ceramic Receivers”. In: *29th SolarPACES Conference*, SolarPACES. 2023.
- [81] Tianju Chen and Mark C. Messner. “A Universal Inelastic Constitutive Model for High Temperature Deformation.” In: *ASME 2023 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2023.
- [82] Tianchen Hu and Mark C. Messner. “What Best Correlates to High Temperature Failure: Strain, Stress, Dissipation, or Something Else?” In: *ASME 2023 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2023.
- [83] Bipul Barua, Mark C. Messner, and T.-L. Sham. “Design Charts for an Integrated Creep-fatigue Damage Evaluation Approach”. In: *ASME 2022 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2022.

- [84] Bipul Barua, Mark C. Messner, and T.-L. Sham. “Nickel Cladded Structural Components for Advanced Reactors”. In: *ASME 2022 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2022.
- [85] F. C. Brust, J. Sallaberry, and M. C. Messner. “High Temperature Flaw Evaluation Code Case: Technical Basis and Examples”. In: *ASME 2022 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2022.
- [86] Pawan Chaugule et al. “Investigating Various Failure Models on Commercial SiC”. In: *28th SolarPACES Conference*. 2022.
- [87] Mark Messner et al. “A Computer Design Tool for Ceramic Receivers”. In: *28th SolarPACES Conference*. 2022.
- [88] Mark C. Messner. “A Viscoplastic Model for Alloy 800H for use with the Section III, Division 5 Design by Inelastic Analysis Methods for Class A Components”. In: *ASME 2022 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2022.
- [89] B. Barua, R. I. Jetter, and T.-L. Sham. “Simplified Criteria with Reduced Testing Effort for Selecting Clad Materials for High Temperature Reactor Structural Components”. In: *ASME 2021 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2021.
- [90] Bipul Barua and Mark Messner. “Fast Heuristics for Receiver Life Estimation and Design”. In: *27th SolarPACES Conference*. 2021.
- [91] David Dewees et al. “Comparison of candidate steady loading elevated temperature design-by-analysis methods”. In: *ASME 2021 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2021.
- [92] Michael McMurtrey and M. Messner. “Qualification Challenges for Additive Manufacturing in High Temperature Nuclear Applications”. In: *ASME 2021 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2021.
- [93] M. C. Messner and T.-L. Sham. “A Viscoplastic Model for Alloy 617 for use with the ASME Section III, Division 5 Design by Inelastic Analysis Rules”. In: *ASME 2021 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2021.
- [94] Andrea Nicolas, Mark Messner, and T.-L. Sham. “A Probabilistic Margin Assessment of the ASME Section III, Division 5 Primary Load Design Rules for Class A Components”. In: *ASME 2021 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2021.
- [95] Andrea Rovinelli, Mark Messner, and T.-L. Sham. “A Comprehensive Comparison between Different Multiaxial Cycle Counting Procedures”. In: *ASME 2021 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2021.
- [96] B. Barua et al. “Acceptance Criteria for the Mechanical Integrity of Clad/Base Metal Interface for High Temperature Nuclear Reactor Cladded Components”. In: *ASME 2020 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2020.
- [97] B. Barua et al. “Development of Design Method for High Temperature Nuclear Reactor Cladded Component”. In: *ASME 2020 Pressure Vessels and Piping Conference*. American Society of Mechanical Engineers. 2020.
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